# anime-rag

Multimodal (Agentic) RAG over the Manga109-s manga dataset. Individual NLP course project.

- **Code:** https://github.com/martin76ec/manga-multi-modal-rag
- **Built index (embeddings + metadata, no images):** https://huggingface.co/datasets/martin76ec/Manga109-s-indexes

## Course project overview

### Topic

**RAG / Agentic RAG.** Not built with LangChain/LangGraph. This is a deliberate, tested deviation, not an oversight:

- `google/gemma-4-26B-A4B-it` (needed for vision, so the agent can judge a panel directly, not just via translated dialogue) does not reliably use native tool-calling. Confirmed with the raw HF Inference API, bypassing LangChain entirely: given a registered tool, it just answered in plain text claiming it had none.
- Wiring it into LangGraph's `create_react_agent` was worse: it hallucinated a complete fake tool call, with a fabricated URL and fictional data.
- Models that do tool-call reliably on the free providers I have access to (`Llama-3.3-70B`, `Qwen2.5-72B`) are text-only. The one vision+tool-calling model I found (`Qwen2.5-VL-32B`) also ignored the tool and hallucinated an answer.

No model available to me does both reliably at once. The hand-built agent in `agent.py` (MRKL-style `Thought → Action → Observation → Final Answer`, plain-text prompted, with retry/degeneration detection added after real provider failures) is the result of testing LangChain and finding it less reliable here, not a workaround for skipping it.

### Problem and data

Manga dialogue in Manga109-s is Japanese-only, and panels carry visual information the transcribed text often doesn't describe. Goal: let someone who doesn't read Japanese ask a question in English and get an answer grounded in cited panels, using both artwork and dialogue.

**Data:** Manga109-s (gated), 87 licensed manga volumes with page images and per-panel annotations. Fully indexed: **87 books, 82,124 panels**.

### Baseline vs. proposed solution

| | Baseline | Proposed |
|---|---|---|
| Retrieval | single modality: text-only or image-only CLIP similarity | blended score ( `0.6 image + 0.4 text` , tunable) over the same embeddings |
| Query language | Japanese only | English or Japanese (NLLB-200 translation bridge) |
| Answering | top-k panels returned directly | MRKL+IRCoT agent: multi-round, vision-grounded, cites book/page, says "I don't know" instead of guessing |

### Comparison methodology and results

No formal labeled benchmark (hit-rate@k, RAGAS) was built for this submission; that's a stated limitation (see Conclusions). Below is a comparison from real development-time queries, each checked against the actual panel image.

**Modality ablation** (query: `"police officer"` ):

| Mode | Top-1 | Score | Verified correct? |
|---|---|---|---|
| Image-only | `AisazuNihaIrarenai_007` | 0.256 | not independently verified, weaker signal |
| Text-only | `AisazuNihaIrarenai_057` | 0.740 | yes, panel reads "ちょっと警察の方!!" |
| Blended (default) | `AisazuNihaIrarenai_057` | 0.403 ( `0.6x0.178 + 0.4x0.740` ) | yes, checked by hand |

Isolating either modality changes the ranking. Text carried more signal here; the blend keeps results robust for queries where the opposite holds (e.g. a purely visual one).

**Baseline vs. agent:**

| Query | Baseline | Agent |
|---|---|---|
| "police officer" | correct top-1, single shot | same, 1 round |
| "What happens when a police officer shows up at someone's house?" | needs the user to already know the right search term | finds and cites the same panel unprompted, with a grounded answer |
| "someone holding a gun" (no real match indexed) | returns a plausible but wrong top result | correctly reports no match found |
| "someone holding a gun" (real match indexed) | finds the panel, no explanation | finds it and visually confirms it ("a Glock handgun"), a detail the text doesn't mention |
| multi-hop father/parenting question | can't answer at all | 4 self-generated retrieval rounds, found the father, then honestly declined to confirm a mistranslated detail instead of guessing |

The agent's real advantage is saying "I don't know" instead of confidently returning a wrong top-1, and chaining queries for questions that need more than one search. That costs more (next section), so it's a trade-off, not a strict upgrade.

## Cost comparison

| | Baseline | Agent |
|---|---|---|
| Where it runs | fully local | retrieval local, reasoning via HF Inference API (network, third party) |
| LLM calls/question | 0 | up to 5 ( `max_steps=4` + forced final) |
| Tokens/call | 0 | `max_tokens=250` , plus growing conversation history |
| Images sent | 0 | up to 3 panels/round x 4 rounds, 384px JPEG |
| Latency | sub-second | tens of seconds |
| Monetary cost | none | HF free tier (hit a 429 rate limit once) |

### Local compute/storage (shared by both modes):

| | Value |
|---|---|
| Panels indexed | 82,124 (87 books) |
| Embedding throughput | ~58 panels/sec (shared RTX 4060 Ti, 8GB) |
| Local models | CLIP Japanese (~790MB), NLLB-200 (~2.4GB), both free |
| Index size | 380.5MB (one Parquet file) |
| Raw data on disk | ~7.8GB |

## Conclusions

**What worked:**

- Blending image and text embeddings changes outcomes versus either alone, verified by hand.
- The translation bridge makes the system usable without Japanese, fully local, no API key.
- The agent's vision is not cosmetic: it found and described a detail ("a Glock handgun") the dialogue never mentioned.
- The agent consistently abstained instead of hallucinating a plausible wrong answer, which matters most for a citation-grounded tool.

**What didn't work, fixed along the way:**

- CLIP collapsed very short dialogue (e.g. "……") into a generic embedding; fixed with a minimum-text filter.
- NLLB dropped key words translating stuttering dialogue; fixed with beam search and repetition controls.
- The HF provider occasionally returned malformed output: once a fake empty tool-call, twice different repetition-loop patterns. Each required real reproduction and root-causing.
- The LangChain/LangGraph attempt failed outright for this model, a real negative result.

**Limits:**

- No formal quantitative benchmark was built; the comparison above is real but small and not randomly sampled.
- Recall, not precision, is the main failure mode: evidence missing from the index simply can't be found.
- Character-name translation is unreliable (no manga-specific dictionary).
- Agent reliability partly depends on third-party provider behavior outside this project's control.

## AI-generated content declaration

Built with extensive assistance from Claude (Anthropic) as an engineering collaborator: architecture, implementation, debugging real failures, and this documentation. Design decisions and acceptance of each change were mine; the AI did the hands-on work under that direction, and its claims were verified against real run output, not taken at face value. Adjust this section for your institution's disclosure policy before submitting.

## Presentation (5 min + 3 min Q&A)

Suggested flow: problem/data (30s), topic and the LangChain/LangGraph finding (45s), live demo of `anime-rag ask` (2 min), comparison and cost tables (1 min), conclusions/limits (45s).

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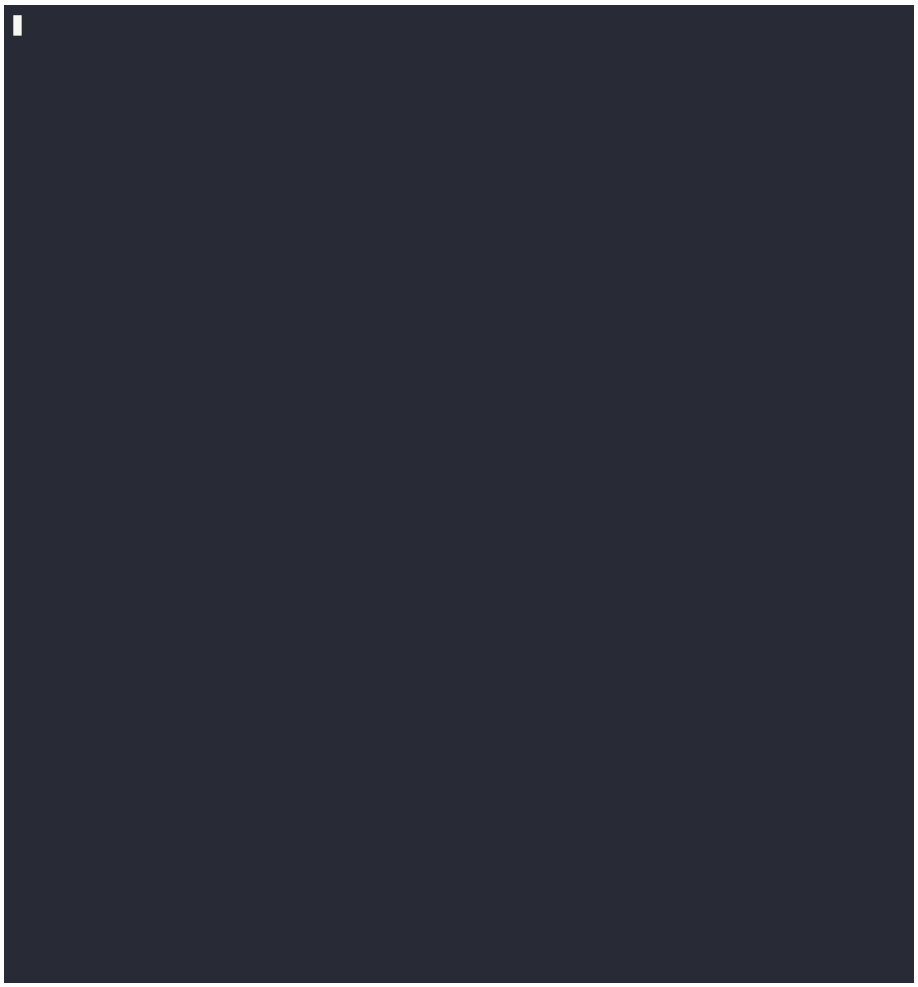
## Setup

```
uv sync
hf auth login                  # needed once, for the gated dataset
```

Manga109-s is a gated dataset. Request access on the dataset page first.

## Usage

```
# interactive REPL: loads models once, then take as many questions as you like.
# menu-driven (search by text / by image / ask the agent / quit), colored
# output, and an inline true-color thumbnail of the top result panel
uv run anime-rag ask
```



## How it works

- `source.py` opens the dataset's zip archive through `HfFileSystem` and extracts only the requested books' images/annotations (partial HTTP range reads).
- `annotations.py` parses each book's Manga109 XML into pages of frames (panels), text boxes, and character face/body boxes.
- `panels.py` crops one image per frame and associates it with the dialogue text and character names whose boxes fall inside that frame.
- `embedder.py` wraps the Japanese CLIP model to embed panel crops and text into a shared space.
- `translate.py` translates English queries to Japanese before embedding, and Japanese result dialogue back to English for display, via `facebook/nllb-200-distilled-600M` (local, no API key needed).
- `indexing.py` builds and persists a per-book index (embeddings + metadata) under `data/index/`.
- `search.py` ranks panels for a query by a weighted blend of image-embedding and text-embedding cosine similarity. Panels whose dialogue is empty or too short to embed meaningfully (e.g. "……") fall back to image-only scoring.
- `agent.py` implements a **MRKL + IRCoT** retrieval agent on top of `search.py`: a MRKL-style explicit `Thought → Action → Observation → Final Answer` transcript (`llm.py`, via the HF Inference API, the only part of this project that's not local) drives a single `search_manga` tool, but IRCoT-style, each retrieval query is the agent's own latest reasoning step rather than a fixed query, so it can chase multi-hop questions across several retrieval rounds and still ground its answer in cited book/page evidence (or admit it doesn't know, instead of guessing). The LLM (`google/gemma-4-26B-A4B-it`) is vision-capable, so each Observation includes the actual panel images, not just translated dialogue; it can judge what's visually happening in a panel even when the dialogue doesn't describe it.

Downloaded data and built indices live under `data/` (gitignored). The dataset license forbids redistribution.

`anime-rag agent` calls the Hugging Face Inference API (model: `google/gemma-4-26B-A4B-it` by default) and needs network access plus an HF token with inference access (`hf auth login`); everything else in this project (embedding, translation, indexing) runs fully locally.

## Indexing the full dataset (e.g. on a bigger GPU)

The dataset has 87 books (82,124 panels total once all are indexed). Embedding that locally on a small/shared GPU takes a while, so on a bigger box (this was written with an H200 in mind):

```
# on the bigger machine: clone this repo, then
uv sync
hf auth login                       # needs Manga109-s access

# index every book in one go; bump --batch-size way up on a big GPU
uv run anime-rag index --all --batch-size 256

# or, as a single script (also pushes the result to HF when given a repo id):
./scripts/h200_index_all.sh yourname/manga109-rag-index
```

`--all` opens the dataset zip once and indexes everything it finds via `anime-rag books`, skipping any book already indexed (safe to re-run / resume after an interruption).

### Moving the index to another machine

`push-index` / `pull-index` upload/download **only the built index** (embeddings + metadata) via a HF dataset repo. It never includes the actual panel crop images, which live in a separate directory ( `data/raw/crops/` ) the license is stricter about: this is derived numeric output, analogous to publishing a pretrained model, not a copy of the dataset itself (see the license note below). Push it **private** (this project's own index, linked above, is public by request).

The index is consolidated into a single `index.parquet` file before uploading. Parquet is the standard single-file format for embeddings + metadata (what HF's own `datasets` library uses), and far easier to move around than one directory per book (87 books = 348 small files otherwise). `pull-index` downloads that one file and transparently re-splits it back into the usual `data/index/<book>/` layout, so nothing else changes.

```
# on the box that built the index
uv run anime-rag push-index yourname/manga109-rag-index

# if you previously pushed with an older version of this tool that uploaded
# the raw per-book folders instead of a single file, add --clean once to
# remove that old structure from the repo after the parquet file is up
uv run anime-rag push-index yourname/manga109-rag-index --clean

# on another machine
uv run anime-rag pull-index yourname/manga109-rag-index

# the pulled index has no crop images yet, regenerate them locally
# (re-extracts raw pages from the dataset zip, crops them; no GPU/model
# inference involved, so it's cheap even without the bigger box)
uv run anime-rag crops
```

Search/citations work immediately after `pull-index` (it's pure embedding math); you only need `anime-rag crops` once you want to actually view a result's panel image or use the vision-capable agent.

## License note

Manga109-s is distributed under its own license (non-commercial-friendly but not open), which caps how much of any one book's pages may be published and forbids redistributing the dataset itself. See the dataset page for details.